

US ECONOMICS ANALYST

From Innovation to Productivity Boom: Lessons from the ICT Revolution for the AI Era

- We expect AI to meaningfully boost productivity growth over the next decade. Already, dozens of research studies and corporate anecdotes show evidence of large productivity gains at the micro level in a range of applications. But key questions for both equity and macro investors are when and where we might first see evidence of AI-driven productivity gains at the macro level.
- We look back at the information and communications technology revolution (ICT) from roughly 1980–2000 to draw insights from history. A key lesson of that period is that the path from technological innovation to broad-based productivity gains is neither fast nor uniform.
- When did the impact of ICT on productivity first appear? The ICT productivity boom arrived 15 years after the commercialization of the personal computer. While ICT investment began rising sharply in the early 1980s, industry-level data show that its impact on productivity followed a J-curve: a modest drag on measured productivity growth in the first four years, with clear gains emerging only after eight years and peaking at year twelve. The impact was slow because key ICT component costs were initially high, network effects did not kick in until adoption reached a critical mass, and human and institutional impediments slowed technology adoption and were only gradually overcome with substantial investment in intangible organizational capital.
- Where did convincing evidence that ICT was boosting productivity first appear? Early productivity gains appeared in industries where ICT was already deeply embedded in the production process even before the innovation breakthrough of the early 1980s, in industries that invested in ICT earliest, and especially in industries that invested heavily in reorganization to use ICT more effectively. While productivity growth in these industries picked up earlier and more strongly than in the aggregate statistics, providing persuasive early evidence that ICT was having a meaningful impact, some of the key measures used today to identify these early winners, such as intangible investment, might not have been obvious to observers at the time.
- To what degree will these historical lessons from ICT apply to AI? With regard to timing, we expect to see a quicker impact of AI on productivity because costs are falling much faster for AI models than they did for ICT equipment, and network effects are less vital for realizing productivity gains. But similar human and institutional obstacles to adoption are again likely to delay the impact on

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productivity. While investment in hardware is already rising even faster than during the ICT buildout, investment in reorganization of work processes appears to be moving more slowly.

- With regard to where to look for the earliest tests of whether AI is boosting productivity growth, the industries to watch are clearer today thanks to the availability of higher frequency data on AI exposure and adoption in addition to data on investment in tangible and intangible capital. Our composite ranking across these measures implies that industries such as information, professional services, insurance, and finance should show the earliest and largest pickups in productivity growth. While productivity trends for these industries should provide the earliest test, industry-level data come out with a lag, so even here the evidence will likely take at least several years to arrive.

From Innovation to Productivity Boom: Lessons from the ICT Revolution for the AI Era

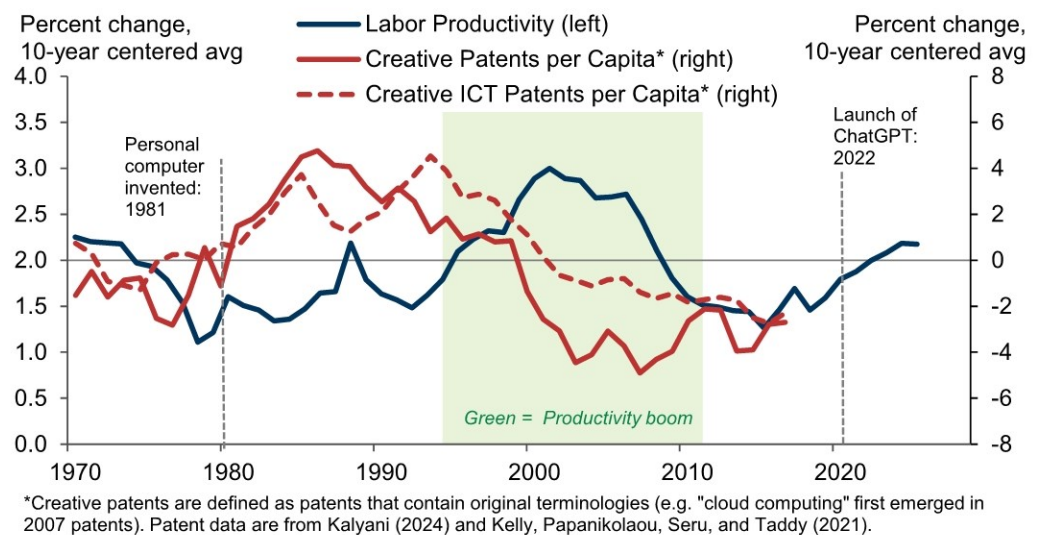
We expect AI to meaningfully boost productivity growth over the next decade. Already, dozens of research studies and corporate anecdotes show evidence of large productivity gains at the micro level in a range of applications. But key questions for both equity and macro investors are when and where we might first see evidence of AI-driven productivity gains at the macro level.

We look back at the information and communications technology revolution (ICT) from roughly 1980–2000 to draw some insight from history. A key lesson of that period is that the path from technological innovation to broad-based productivity gains is neither fast nor uniform.

When Did the Impact of ICT on Productivity First Appear in the Macro Data?

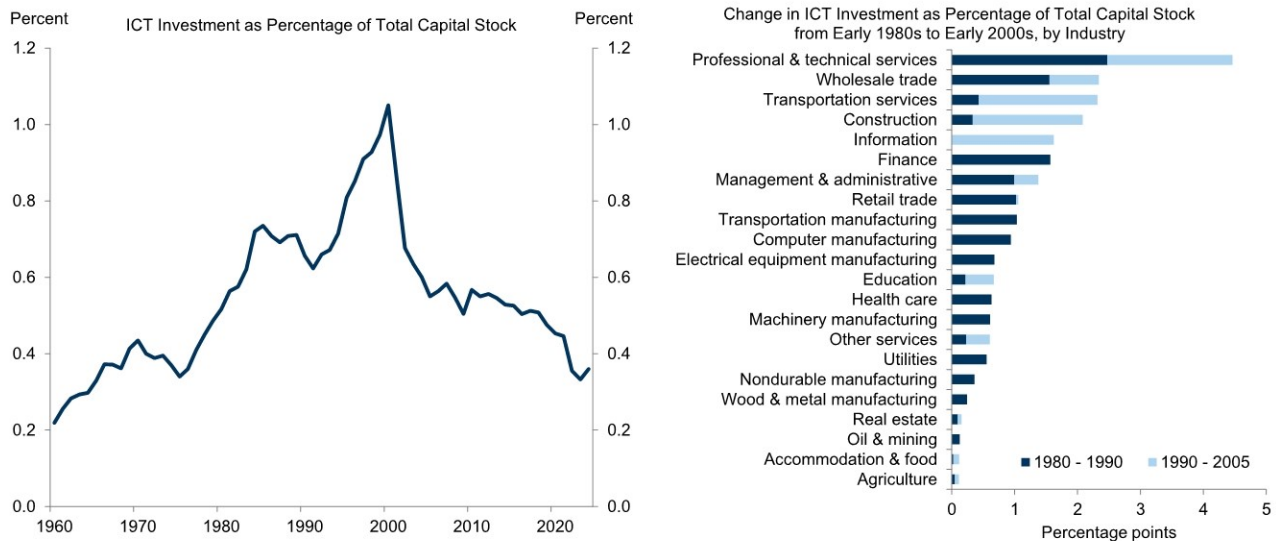
The commercialization of the personal computer in the early 1980s set off a wave of technological innovation. Exhibit 1 shows that there was a sharp and sustained rise in breakthrough ICT patents per capita through the late 1980s and early 1990s. But productivity growth remained stubbornly flat for nearly 15 years, accelerating only in the second half of the 1990s.

Exhibit 1: The Commercialization of the Personal Computer in the Early 1980s Set Off a Wave of Innovation, but Productivity Growth Did Not See a Meaningful Increase Until the Late 1990s



Source: Goldman Sachs Global Investment Research, Department of Labor

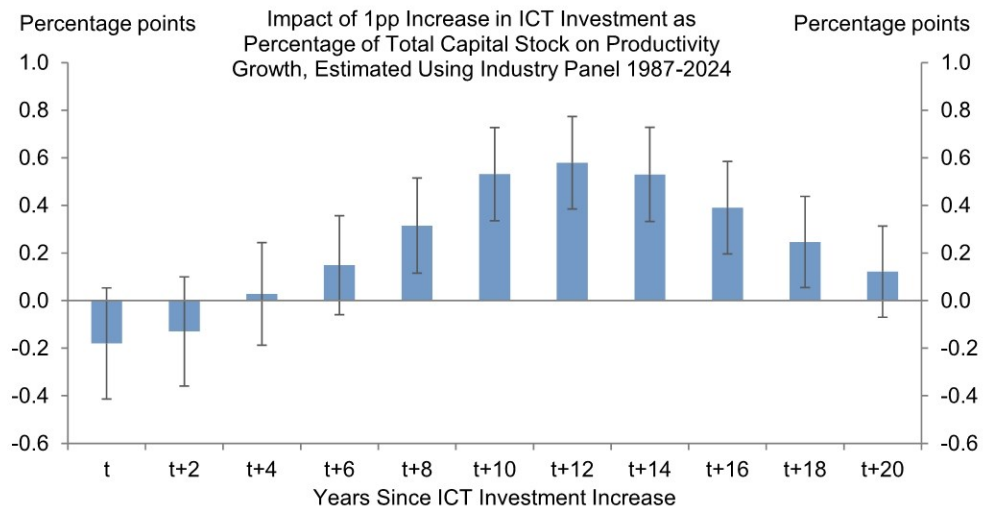
The impact on productivity growth took a while to appear even though ICT investment began to rise sharply as early as the 1980s (Exhibit 1, left). The increase in ICT investment was fairly broad-based, though industries like professional services, wholesale trade, transportation, and finance invested much more intensively (Exhibit 2, right).

Exhibit 2: ICT Investment Began Rising Sharply in the Early 1980s, With Increases Seen Across Most Industries


Note: ICT investment includes investment in personal computers, mainframes, storage devices, printers, terminals, tape drives, system integrators, and communication devices. Data are from BEA's fixed asset survey.

Source: Goldman Sachs Global Investment Research, Department of Commerce

But it took a long time for an increase in ICT investment to have a notable impact on productivity. Using industry panel data, we find that an increase in ICT investment actually led to a modest decline in measured productivity growth in the first four years. The positive effects emerged only with a substantial lag: a 1pp increase in ICT investment as a percentage of the capital stock did not produce a statistically significant rise in productivity growth until roughly 8 years later, with the impact peaking at around 0.6pp in year twelve.¹

Exhibit 3: ICT Investment Initially Weighed on Productivity Growth, With Positive Effects Emerging Roughly 8 Years Later


Note: Error bars indicate 95% confidence interval.

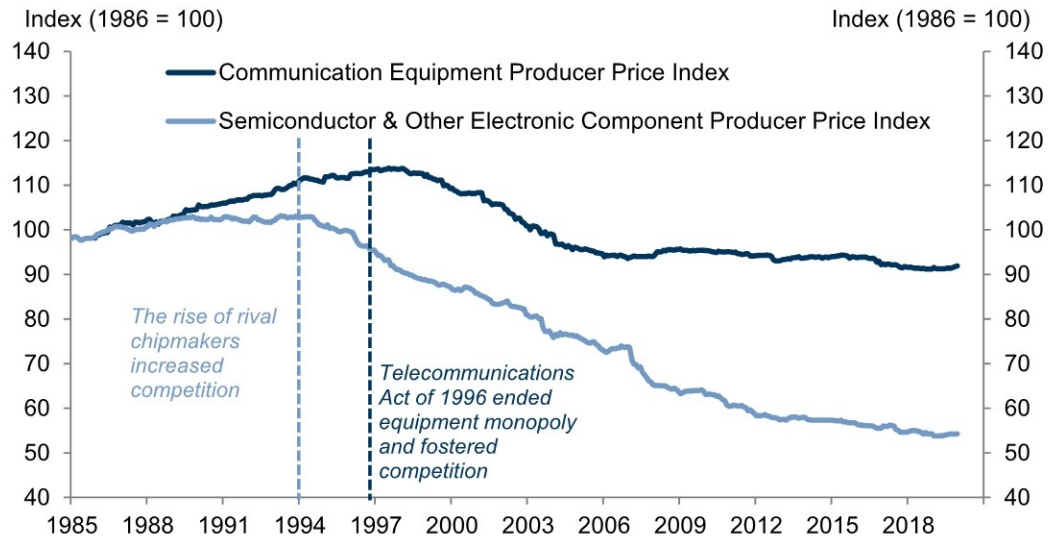
Source: Goldman Sachs Global Investment Research

We see three factors that delayed productivity gains.

¹ These effects are similar in magnitude to those from [academic studies](#) that estimated the long-run impact of ICT investment on productivity growth at the industry level.

First, key ICT components—such as semiconductors and telecommunication devices—remained expensive throughout the 1980s. Prices only began falling after regulatory interventions and increased competition opened previously concentrated markets in the 1990s.

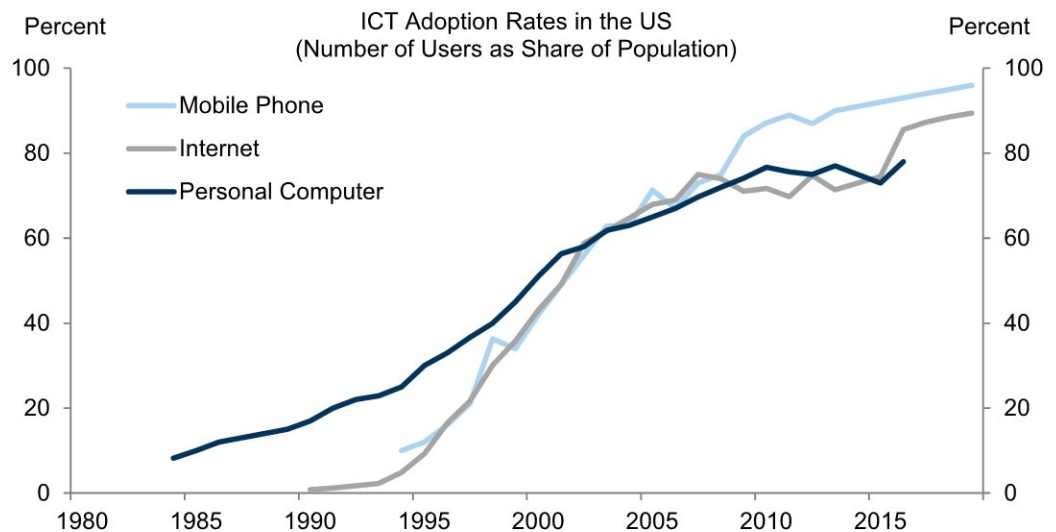
Exhibit 4: Key ICT Components Remained Expensive Through the 1980s; Prices Only Started to Fall Following Increased Competition and Regulatory Changes in the 1990s



Source: Goldman Sachs Global Investment Research, Department of Labor

Second, many ICT applications such as the internet exhibited strong network effects, generating far greater value for their users after adoption reached a critical mass. Productivity gains tended to emerge only after adoption rates passed an inflection point in the late 1990s.

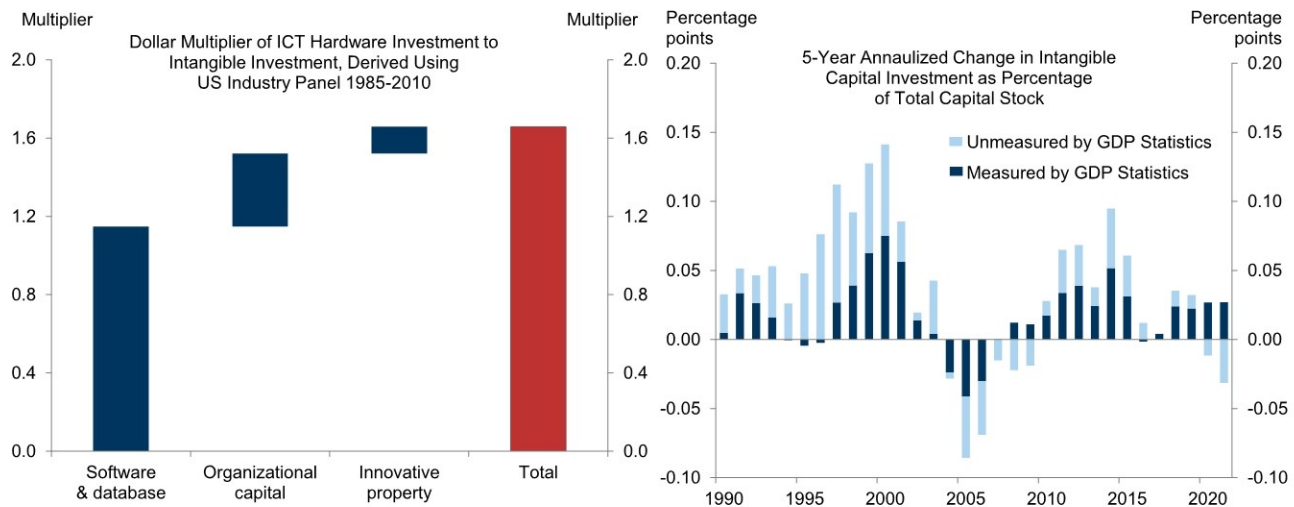
Exhibit 5: ICT Applications Exhibited Strong Network Effects, and Productivity Gains Only Emerged After Adoption Rates for Technologies Like the Internet Passed an Inflection Point in the Late 1990s



Source: Goldman Sachs Global Investment Research

Third, an even more important bottleneck was the time and resources required for firms to build intangible assets needed to support the technological transition, such as redesigning workflows, retraining workers, and restructuring organizations, and building new data systems, as discussed in detail in our Global Economics team's prior report. The left panel of Exhibit 6 shows that each \$1 of ICT hardware investment likely required at least an additional \$1.7 of complementary intangible investment, with roughly two-thirds directed toward software and data systems and the remainder toward workforce reorganization.² The right panel of Exhibit 6 shows that there was a clear increase in intangible capital investment in the mid-1990s, and even this likely understates the increase because much of the early spending for reorganization was not captured in the official GDP statistics.

Exhibit 6: ICT Required Meaningful Investment in Intangible Capital to Reach Its Full Productivity Potential, and Companies Only Started to Invest Heavily in Intangible Capital in the Late 1990s



Source: Goldman Sachs Global Investment Research, EUKLEMS, Haver Analytics

Where Did the Impact of ICT on Productivity First Become Visible?

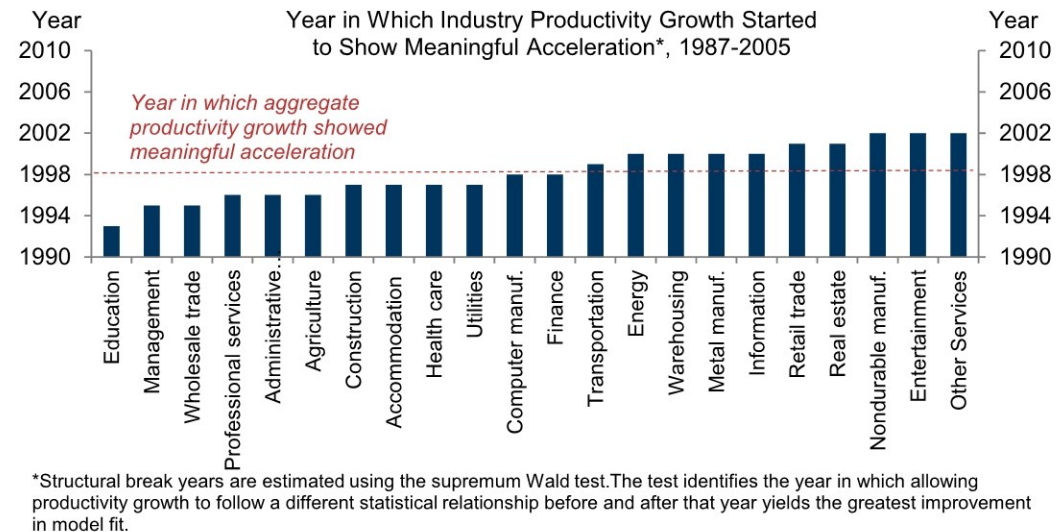
To identify where the ICT productivity boom first became visible at the industry level, we use a standard statistical test for structural breaks in productivity growth across major industries. Exhibit 7 shows the year in which each industry first exhibited a statistically significant acceleration in productivity growth. We find that education, management, wholesale trade, professional services, and administrative support were among the earliest to show a clear pickup, in some cases several years ahead of the acceleration in the aggregate data.

How confident can we be that these structural breaks were driven by ICT rather than other factors? As a check, we estimate the impact of ICT investment on productivity growth separately for each major industry group and combine these estimates with actual ICT investment data at the detailed industry level to estimate impulses from ICT investment to productivity growth for each industry. We then run the same structural break test on these industry-level ICT impulses to productivity growth and find that the resulting break years are broadly consistent with those identified using productivity data

² These estimates are broadly in line with our Global Economics team's prior estimates using country-level panel but are on the conservative side compared to estimates from academic literature.

alone, supporting the conclusion that these early productivity pickups were most likely ICT-driven.

Exhibit 7: Some Industries Showed a Clear Pickup in Productivity Growth Several Years Ahead of the Acceleration in the Aggregate Data

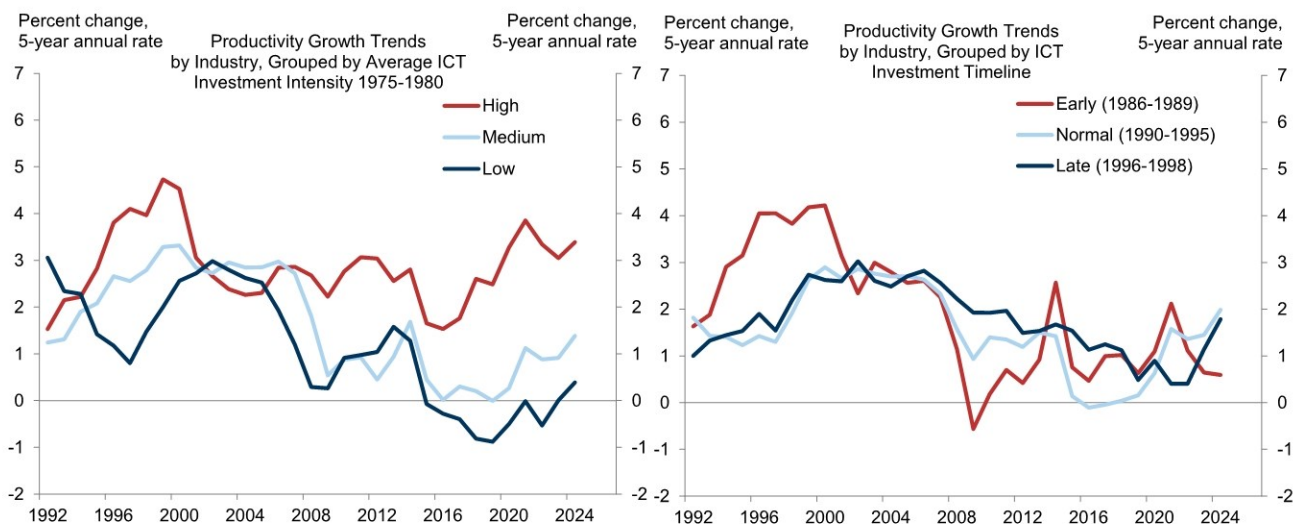


Source: Goldman Sachs Global Investment Research, Department of Labor

What characteristics distinguished the earliest and largest beneficiaries of ICT?

We find that industries where ICT was already deeply embedded in the production process before the technological breakthroughs of the early 1980s—or those that were among the earliest to invest heavily in the new technologies when they first became available—tended to see an earlier productivity payoff, with productivity growth in these industries peaking as early as the mid-1990s (Exhibit 8).

Exhibit 8: Industries Where ICT Was Already Deeply Embedded in the Production Process Before the Technological Breakthrough of the Early 1980s or Those That Adopted the New Technologies Earliest Tended to See an Earlier Productivity Payoff

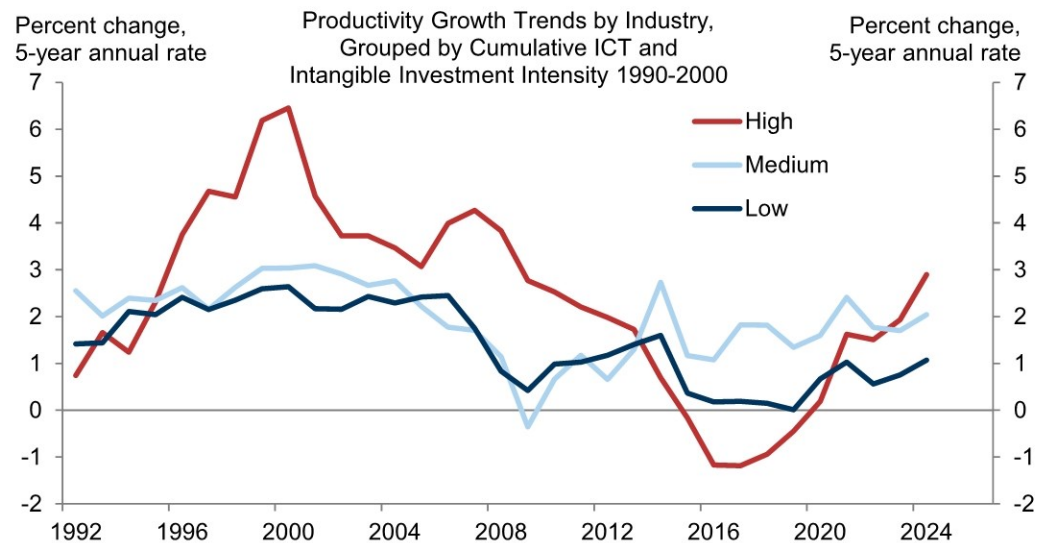


Source: Goldman Sachs Global Investment Research, Department of Labor

But the ultimate winners were those that invested heavily in intangible organizational

capital alongside their ICT investment—restructuring workflows, retraining workers, and redesigning business processes. As Exhibit 9 shows, these industries went from underperforming the rest of the economy in productivity growth to pulling sharply ahead by the early 1990s—and sustained that lead for years afterward. This finding reinforces a central lesson of the ICT experience: the industries that gained the most were not simply those that adopted the technology first or invested the most in hardware, but those that also made the largest complementary investments in reorganizing their operations around it.

Exhibit 9: The Ultimate Industry Winners Were Those That Invested Heavily in Reorganization Capital Alongside Their ICT Investment

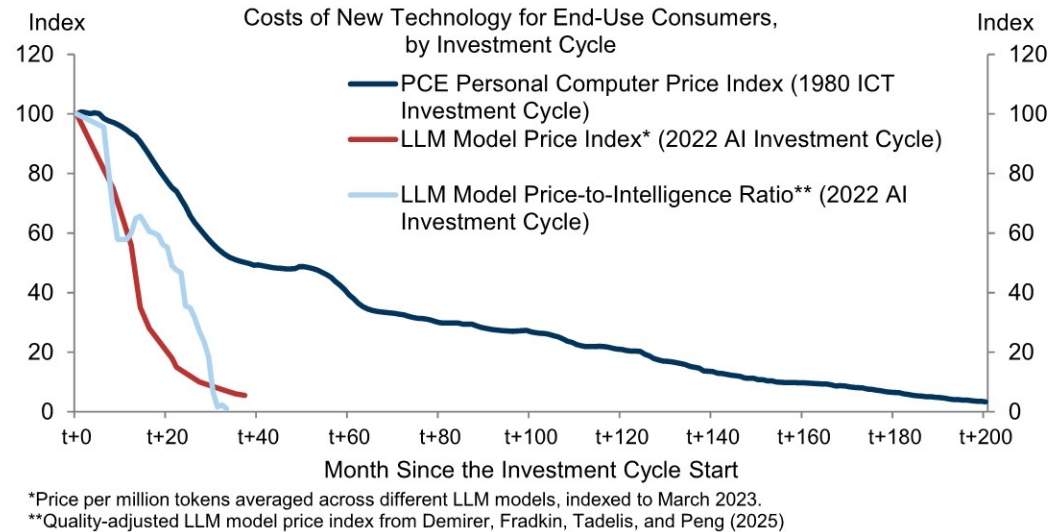


Source: Goldman Sachs Global Investment Research, Department of Labor

The Lessons of the ICT Revolution for the AI Era

To what degree will these historical lessons from ICT apply to AI? With regard to timing, we expect to see a quicker impact of AI on productivity because costs are falling faster for AI models than they did for ICT equipment. That said, headline pricing for some frontier US AI models has risen in recent months as leading providers seek to expand margins. Our equity analysts expect overall token prices to stabilize as competitive pressure and continued declines in underlying compute costs limit providers' ability to sustain elevated pricing. In addition, unlike with communication-oriented technologies, network effects are probably less vital for realizing productivity gains from AI.

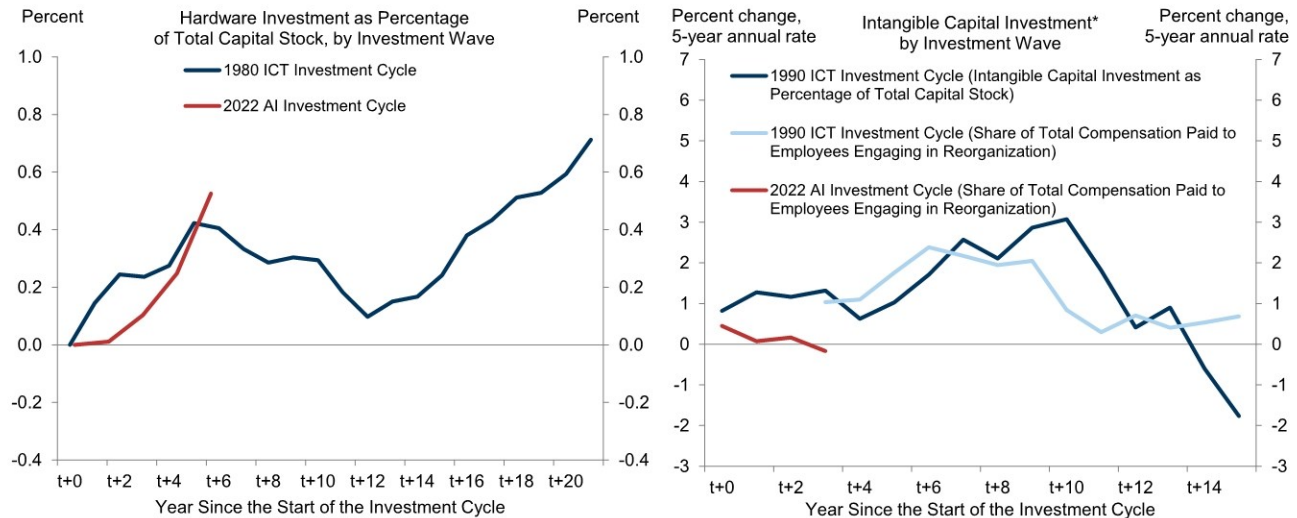
Exhibit 10: The Price of Using AI Models Is Falling Much More Quickly Than Personal Computer Prices Did



Source: Goldman Sachs Global Investment Research, Department of Commerce

But similar human and institutional obstacles to adoption are again likely to delay the impact on productivity. The left side of Exhibit 11 shows that investment in hardware is already rising even faster than during the ICT buildout. Investment in reorganizing work processes—as measured by the share of compensation paid to employees engaging in reorganization—appears to be moving more slowly (Exhibit 11, right), though some intangible investment already underway might not be fully captured by official measures. For example, a recent Atlanta Fed survey of enterprises implies roughly \$280 billion in AI-related intangible capital spending in 2026, and [our prior analysis](#) based on company data suggest that labor costs associated with the AI transition are likely running at \$150bn/year in the US, and executive time allocations suggest \$40bn/year in organization capital investment is likely underway.

Exhibit 11: Investment in Hardware Supporting AI Is Already Growing Very Quickly, but the Investment in Intangible Capital That Will Likely Be Required to Fully Exploit AI Appears Not to Have Followed Yet



*EUKLEMS intangible investment data only starts from 1987, so we also track intangible investment using share of total compensation paid to employees engaging in reorganization activities from the CPS data. To identify employees engaging in reorganization activities, we follow the method proposed by Squicciarini and Le Mouel (2012).

Source: Goldman Sachs Global Investment Research, Department of Labor, Department of Commerce

With regard to where to look for the earliest tests of whether AI is boosting productivity growth, the industries to watch are clearer today thanks to the availability of data on AI exposure and adoption in addition to data on investment in tangible and intangible capital. Our composite ranking across these measures implies that industries such as information, professional services, insurance, and finance should show the earliest and largest pickups in productivity growth.

Exhibit 12: We Classify Industries by Their Potential for an Early Impact of AI on Productivity Growth Based on Pre-Existing IT Exposure, AI Adoption, AI Exposure, and Investment in Work Reorganization

AI Potential Early Impact Score						
	Industry Name	Pre-Exposure to Information Technology*	AI Adoption Rate** (May 2026)	AI Exposure Score***	Reorganization Intensity Since 2022^	Average Score
High Potential	Information & data processing	4.2	1.9	1.8	0.0	1.97
	Professional services	2.4	2.0	1.4	0.1	1.48
	Motion picture & sound	1.5	1.9	1.0	0.4	1.21
	Insurance carriers	1.9	0.7	2.5	-0.7	1.09
	Credit intermediation	0.4	0.8	1.9	0.9	0.98
	Computer and electronics manufacturing	2.1	0.8	0.7	0.2	0.97
	Administrative & support	1.7	0.4	0.3	0.5	0.74
	Electrical equipment manufacturing	0.5	0.6	0.4	1.3	0.70
	Management of companies	0.4	-1.5	1.3	2.3	0.64
	Miscellaneous manufacturing	0.5	0.1	0.7	0.9	0.56
	Broadcasting & telecom	0.9	0.8	0.6	-0.3	0.48
	Machinery manufacturing	0.3	0.3	0.4	1.0	0.48
	Chemical products	1.1	1.2	-0.2	-0.2	0.48
	Warehousing & storage	-0.4	1.8	-0.9	1.1	0.38
	Plastics & rubber products manufacturing	-0.2	-0.3	0.1	2.0	0.37
Medium Potential	Real estate	-0.9	1.3	0.2	0.8	0.35
	Printing & related support activities	-0.2	0.9	0.8	-0.1	0.34
	Rental & leasing services	-0.5	0.4	0.2	1.1	0.28
	Transportation equipment manufacturing	0.9	-0.4	0.5	0.0	0.27
	Educational services	-0.5	1.4	-0.3	0.3	0.23
	Wholesale trade	0.1	1.1	0.3	-0.7	0.20
	Performing arts	0.3	1.0	0.0	-0.5	0.20
	Nonmetallic mineral	0.0	0.0	-0.6	1.1	0.14
	Ambulatory health care	-0.4	0.8	-0.1	0.0	0.07
	Accommodation	-0.7	-0.6	0.7	0.6	-0.01
	Paper products manufacturing	-0.4	1.2	-0.3	-0.5	-0.01
	Furniture and related products manufacturing	0.0	0.0	0.5	-0.6	-0.03
	Apparel & leather manufacturing	0.2	-0.7	1.6	-1.3	-0.04
	Food & beverage manufacturing	-0.5	0.3	-0.1	0.1	-0.06
	Social assistance	-0.5	0.7	-0.1	-0.3	-0.06
	Transit & ground passenger transportation	-0.7	-0.2	0.1	0.6	-0.06
	Amusements	-0.6	0.0	-0.2	0.4	-0.11
	Food services	-0.6	-0.8	0.9	-0.1	-0.14
	Retail trade	-0.3	-0.4	0.9	-0.9	-0.17
	Oil	-0.8	0.0	-0.3	0.2	-0.22
	Other services	-0.3	-0.5	-0.1	-0.1	-0.25
	Primary metals	-0.5	-1.5	-0.3	1.1	-0.29
	Wood manufacturing	-0.7	-0.5	-0.1	0.1	-0.31
	Other transportation & support activities	-0.2	-0.8	-0.3	0.0	-0.31
	Waste management	-0.8	-0.6	-1.1	0.9	-0.39
	Fabricated metal manufacturing	-0.4	-0.6	0.1	-0.8	-0.42
Low Potential	Truck transportation	-0.6	-0.8	-1.2	0.8	-0.46
	Utilities	-0.8	-0.6	-0.6	0.1	-0.47
	Air transportation	-0.7	-1.5	-1.6	1.4	-0.61
	Hospitals, nursing & residential care	-0.6	-0.7	-0.8	-0.5	-0.63
	Textile mills manufacturing	-0.5	-0.7	1.0	-2.4	-0.66
	Construction	-0.6	-0.5	-1.9	-0.1	-0.77
	Pipeline transportation	-0.8	-1.5	-1.0	0.0	-0.83
	Petroleum & coal products manufacturing	-0.1	-1.5	-1.0	-1.2	-0.94
	Forestry	-0.8	-0.7	-0.3	-2.2	-0.99
	Support activities for mining	-0.7	-0.4	-2.0	-1.3	-1.09
	Water transportation	-0.4	-1.5	-2.0	-1.6	-1.38
	Railroad transportation	-0.8	-1.5	-1.5	-1.7	-1.39
	Mining	-0.8	-1.5	-1.7	-2.0	-1.51

Note: The table shows z-scores of different AI-related measures.

*Software and ICT investment as share of total capital stock in 2019.

**Industry AI adoption rate from Census Business Trends and Outlook Survey.

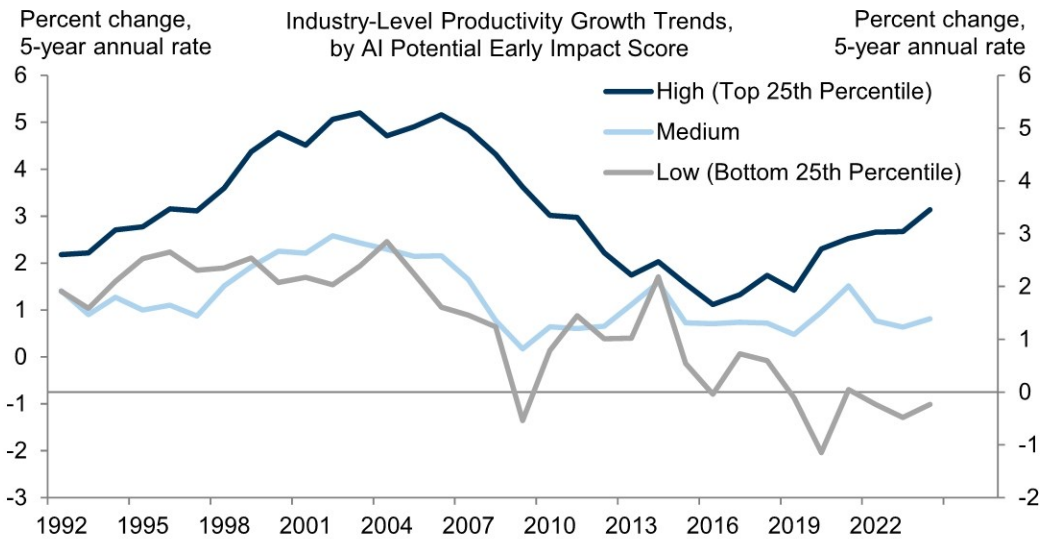
***Share of workers that perform work tasks that can be automated by AI.

^Change in the share of compensation for employees engaging in reorganization activities since 2022.

Source: Goldman Sachs Global Investment Research

Potential evidence of an effect of AI on industry-level productivity growth should be judged against prior trends, and Exhibit 13 shows that productivity growth has already been higher in recent decades in industries that are likely to be the earliest and largest beneficiaries of AI. While trends for these industries should provide the earliest test of whether AI is having a macroeconomically meaningful impact on productivity growth, industry-level data are published with a considerable lag, so even here the evidence will likely take at least several years to arrive.

Exhibit 13: Potential Evidence of AI Effects Should Be Judged Against Prior Trends, and Productivity Growth in Industries Likely to Be the Earliest Beneficiaries of AI Has Already Outperformed Recently



Source: Goldman Sachs Global Investment Research, Department of Labor

Elsie Peng

The US Economic and Financial Outlook

THE US ECONOMIC AND FINANCIAL OUTLOOK

(% change on previous period, annualized, except where noted)

	2023	2024	2025	2026	2027	2025 Q4	Q1	2026 Q2	Q3	Q4	Q1	2027 Q2	Q3	Q4
OUTPUT AND SPENDING														
Real GDP	2.9	2.8	2.1	2.2	2.1	0.5	2.1	2.2	1.9	1.9	2.0	2.1	2.2	2.3
Real GDP (annual=Q4/Q4, quarterly=yoy)	3.4	2.4	2.0	2.0	2.1	2.0	2.7	2.3	1.7	2.0	2.0	2.0	2.1	2.1
Consumer Expenditures	2.6	2.9	2.6	1.7	1.8	1.9	0.5	1.7	1.5	1.5	1.9	2.0	2.1	2.2
Residential Fixed Investment	-7.8	3.2	-2.2	-4.5	1.4	-1.7	-7.8	-6.4	1.5	1.5	2.2	2.2	2.2	2.2
Business Fixed Investment	7.3	2.9	4.1	6.8	5.8	2.4	10.6	7.8	7.5	6.9	4.9	4.9	4.9	4.9
Structures	16.7	1.1	-5.3	-3.6	2.0	-6.6	-4.7	0.4	-1.0	-0.2	3.5	3.5	3.5	3.5
Equipment	2.9	3.5	8.3	10.6	7.1	4.3	15.8	14.7	11.0	9.5	5.0	5.0	5.0	5.0
Intellectual Property Products	6.2	3.5	5.6	8.5	6.2	5.4	13.8	5.0	8.2	7.8	5.5	5.5	5.5	5.5
Federal Government	3.3	3.8	-1.2	-1.4	0.8	-16.7	9.4	-2.0	1.0	1.0	1.0	1.0	1.0	1.0
State & Local Government	3.6	3.8	2.5	1.4	0.9	1.5	1.6	0.8	0.5	1.0	1.0	1.0	1.0	1.0
Net Exports (\$bn, '17)	-925	-1,033	-1,091	-1,084	-1,173	-969	-1,002	-1,086	-1,113	-1,134	-1,150	-1,165	-1,181	-1,197
Inventory Investment (\$bn, '17)	47	44	29	44	65	-16	-17	65	65	65	65	65	65	65
Nominal GDP	6.7	5.3	5.0	6.0	4.7	4.2	5.8	8.7	3.8	4.1	4.7	4.7	4.5	4.3
Industrial Production, Mfg.	-1.0	-1.0	0.8	3.1	3.8	-3.5	4.4	7.7	4.8	3.9	3.3	3.2	3.2	3.3
HOUSING MARKET														
Housing Starts (units, thous)	1,421	1,370	1,356	1,320	1,322	1,323	1,418	1,290	1,280	1,292	1,304	1,316	1,328	1,340
New Home Sales (units, thous)	666	684	679	653	644	711	623	690	662	638	626	632	649	670
Existing Home Sales (units, thous)	4,103	4,067	4,076	4,095	4,254	4,157	4,053	4,073	4,107	4,149	4,191	4,233	4,275	4,317
Case-Shiller Home Prices (%yoy)*	5.3	3.8	0.7	0.7	2.2	0.7	0.3	0.8	0.7	0.7	1.1	1.4	1.8	2.2
INFLATION (% ch, yr/yr)														
Consumer Price Index (CPI)**	3.3	2.9	2.7	3.3	2.2	2.7	2.7	3.9	3.4	3.3	3.0	2.0	2.2	2.2
Core CPI **	3.9	3.2	2.6	2.6	2.2	2.7	2.5	2.8	2.5	2.6	2.5	2.2	2.2	2.2
Core PCE** †	3.1	3.0	3.0	3.0	2.2	2.9	3.1	3.4	3.2	3.1	2.6	2.3	2.3	2.2
LABOR MARKET														
Unemployment Rate (%)^	3.8	4.1	4.4	4.4	4.3	4.4	4.3	4.3	4.3	4.4	4.4	4.4	4.3	4.3
U6 Underemployment Rate (%)^	7.2	7.6	8.4	8.2	8.1	8.4	8.0	8.1	8.2	8.2	8.2	8.2	8.1	8.1
Payrolls (thous, monthly rate)	210	122	10	72	50	-39	73	130	40	45	50	50	50	50
Employment-Population Ratio (%)^	60.1	59.9	59.7	59.1	59.0	59.7	59.2	59.2	59.1	59.1	59.0	59.0	59.0	59.0
Labor Force Participation Rate (%)^	62.5	62.5	62.4	61.8	61.6	62.4	61.9	61.8	61.8	61.8	61.7	61.7	61.7	61.6
Average Hourly Earnings (%yoy)	4.4	3.9	4.0	3.9	3.8	4.0	4.0	4.0	3.9	3.8	3.8	3.8	3.8	3.8
GOVERNMENT FINANCE														
Federal Budget (FY, \$bn)	-1,694	-1,833	-1,775	-1,950	-2,050	--	--	--	--	--	--	--	--	--
FINANCIAL INDICATORS														
FF Target Range (Bottom-Top, %)^	5.25-5.5	4.25-4.5	3.5-3.75	3.5-3.75	3-3.25	3.5-3.75	3.5-3.75	3.5-3.75	3.5-3.75	3.5-3.75	3.5-3.75	3.25-3.5	3.25-3.5	3-3.25
10-Year Treasury Note^	3.88	4.58	4.18	4.40	4.25	4.18	4.30	4.50	4.45	4.40	4.35	4.30	4.25	4.25
Euro (€/€)^	1.11	1.04	1.17	1.18	1.20	1.17	1.15	1.16	1.19	1.18	1.19	1.20	1.20	1.20
Yen (\$/¥)^	141	157	157	158	141	157	159	158	158	158	156	141	101	141

* Weighted average of metro-level HPIs for 381 metro cities where the weights are dollar values of housing stock reported in the American Community Survey. Annual numbers are Q4/Q4.

** Annual inflation numbers are December year-on-year values. Quarterly values are Q4/Q4.

† PCE = Personal consumption expenditures. ^ Denotes end of period.

Note: Published figures in bold.

Source: Goldman Sachs Global Investment Research.

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